

To find the following ML regression method using R2 value.

1. Multiple Linear Regression.

R2 value = 0.9358

2. Support Vector Machine

S.NO	Hyper Parameter	Linear (r value)	RBF (r value)	Poly (r value)	Sigmoid (r value)
1.	C0.1	0.93752	-0.05746	-0.05682	-0.05748
2.	C10	-2.4372	-0.05580	0.02531	-0.05761
3.	C100	-357.0795	-0.03023	0.46566	-0.05878
4.	C1000	-36014.0205	0.16060	0.64032	-0.07070

SVM Regression R2 value = Linear -> C = 0.1 = 0.93752

3. Decision Tree

S.NO	Criterion	Splitter	Max Features	R value
1.	squared_error	Best	None	0.91804
	squared_error	Best	int	0.48968
	squared_error	Best	float	0.91346
	squared_error	Best	sqrt	0.77554
	squared_error	random	None	0.72491
	squared_error	random	int	0.33261
	squared_error	random	float	0.90981
	squared_error	random	sqrt	-0.12476
2.	friedman_mse	Best	None	0.92303
	friedman_mse	Best	int	-0.10394
	friedman_mse	Best	float	0.92760
	friedman_mse	Best	sqrt	0.77872
	friedman_mse	random	None	0.83337
	friedman_mse	random	int	-0.20887
	friedman_mse	random	float	0.88374
	friedman_mse	random	sqrt	0.70436
3.	absolute_error	Best	None	0.96016
	absolute_error	Best	int	-0.33399
	absolute_error	Best	float	0.96355
	absolute_error	Best	sqrt	0.14384
	absolute_error	random	None	0.92489
	absolute_error	random	int	-0.74146

	absolute_error	random	float	0.90033
	absolute_error	random	sqrt	-0.71541
4.	poisson	Best	None	0.94680
	poisson	Best	int	-0.40458
	poisson	Best	float	0.93077
	poisson	Best	sqrt	-0.80423
	poisson	random	None	0.85346
	poisson	random	int	0.78121
	poisson	random	float	0.86679
	poisson	random	sqrt	0.19044

Decision Tree –

Criterion – “absolute_error“, Splitter - ”Best“, Max Features – “Float”, R Value – “0.96355”

Based on the R Value the Best Model to Choose is Decision Tree